FIELD MANUAL

# Engulfs

How to play, the eight skills a round is won with, and the technique behind them.

| | |
|---|---|
| Arena | 8,800 x 8,800 units, 4,100 orbs, 90 viruses |
| Round | 10 minutes, top 5 are paid |
| Field | 100 players in Standard, 50 in High stakes |
| Stake | 1.00 or 2.00 USDC, or play free in Practice |
| Progress | XP and levels for winning; a rating that seats you with players like you |
| Bots | Easy, Normal or Hard, your choice in Practice |

Every figure in this manual is read straight from the simulation the server runs. Where a number appears, that is the number in force.

# The game in ninety seconds

You are a cell. You eat orbs to grow, you eat players smaller than you, and players larger than you eat you. The round runs for 10 minutes. When the whistle goes, everyone still alive is ranked by mass and the top 5 are paid.

## How a round runs

ONE CYCLE

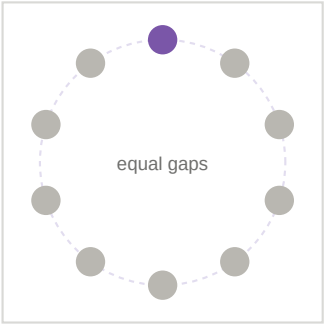
| Lobby | Count 5s | Hold 3s | Round 10 min | Standings 5s | Lobby |
|---|---|---|---|---|---|
| until enough are ready | everyone ready | on the ring, still | the only part that pays | who finished where | ready up again |

When the count runs out, everyone is dealt onto the starting ring and held there for 3 seconds, counted down on screen: nobody can move, so everyone sees the whole opening first, and everyone is let go on the same tick. At the whistle the finishing positions go up for 5 seconds, then everyone goes back to the lobby and nobody is ready: each round, and each stake, is a fresh choice. Finish in the top 5 and a congratulations card shows what the round made you.

### You start on a ring

Every player opens on one circle at equal spacing: the same distance to either neighbour, the same distance to the centre, the same distance to the wall. Nobody is handed a corner to themselves and nobody opens inside somebody's mouth. In a full Standard lobby that gap is about 267 units.

The ring is turned by a fresh angle every round, so there is no memorised opening to learn.




## Four things end your round

- **You are eaten.** Everything you were carrying goes to whoever ate you.
- **You touch a virus while over 126 mass.** You burst into 9 pieces and become food for everyone nearby.
- **The zone closes over you.** From 1:30 the board shrinks; outside it you drain away (see the playbook).
- **The clock runs out.** The only ending that pays.

> **The one rule that decides everything else**
>
> You can eat another cell only when you are at least **9.8% bigger** than it. Not a little bigger - 9.8%. Against a rival of mass 100 you need 109.8. Everything in this manual is downstream of that number: who you chase, who you run from, and how close you dare sit.

## What is in here


| | | | |
|---|---|---|---|
| Controls | Four inputs, and the HUD | The wager | What is at stake, and when |
| The rules | Eating, speed, decay | Eight skills | The habits a round is won with |
| The board | Spawns, orbs and virus layout | Technique | How the work is actually done |
| Splitting | What it buys and what it costs | Playbook | A shape for ten minutes |
| Ejecting | The only thing worth spending on | Reference | Every number on one page |
| Viruses | Cover, mine, and weapon | | |

# Controls

Four inputs. The difficulty is entirely in when you use them.

| ACTION | KEYBOARD | TOUCH | WHAT IT DOES |
|---|---|---|---|
| Move | Mouse | Drag | Your cells steer toward the pointer, faster the further away it is: full speed once it is 60% of the way from the middle of the screen to its nearest edge. Hold it on your cell to stop. |
| Split | Space | Split | Every cell splits in half and the new halves are launched forward. Up to 16 cells. |
| Eject | W | Feed | Throws 13 mass out of the front of each cell, at a cost of 16. |
| Spectate | - | Arrows | Once you are dead, watch anyone still alive. |

## What the screen is telling you

| READOUT | WHY YOU CARE |
|---|---|
| Clock, top left | Time left. It turns red for the last 30 seconds, which is the only moment placing matters more than growing. |
| Rank, top left | Your position and the size of the field. Above 5 is the whole game. |
| Leaderboard, top right | The top ten, with a line ruled under 5th. That line is the pay boundary - watch who is sitting on it. |
| At risk / mass value | What you staked, and what your mass is worth (200 mass is 1.00 USDC). Finish in the top 5 and the mass value at the whistle is what you are paid. Wagered rounds only. |
| Zone call, top centre | The count before the closing zone moves, at 1:40 left; then a red warning for as long as any of your cells is outside it. |
| Minimap, bottom left | Where you are in the arena. Mostly it tells you how close you are to a wall, which is where players get cornered. Late in a round it also draws the zone's final square, and shades what is already outside. |

## What you are actually looking at

Your own cell is predicted locally, so it answers the mouse immediately and does not wait for the server. Everyone else is drawn from what the server last sent, which on a slow connection is a fraction of a second in the past.

The practical consequence: **other players are always slightly further along than they look**. On a contested chase, aim where a target is going rather than where it is drawn. This is also why a fight that felt unfair sometimes was - not because anything is broken, but because you were both acting on slightly different pictures.

## Practice first

The Practice tier runs entirely in your own browser against bots. There is no server, no lobby and no money. Every mechanic in this manual behaves identically there, so it is the right place to learn the timing of a split and the feel of feeding a virus - and, because nothing is predicted over a network, the right place to learn what a clean read looks like. Choose how hard the bots play - Easy, Normal or Hard - under the Practice tier; see *Playing against bots*.

# The rules that never change

Three mechanics shape every decision you make. Learn the numbers and most of the game becomes arithmetic you can do at a glance.

## 1. The 9.8% rule

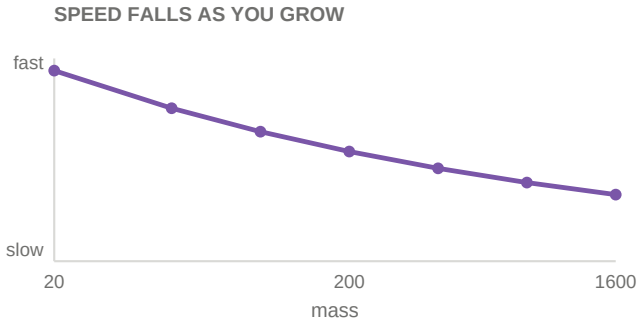



Eating a player pays all of their mass, so players are worth far more than orbs. An orb is 4 mass, and it takes about 20 of them to go from your opening 20 to 100.

## 2. Mass costs speed

Speed falls as mass$^{-0.24}$. Doubling your mass costs you about 15% of your top speed, and it never stops costing. The biggest cell on the board is also the one that cannot catch anything and cannot run away.

This is why the leader is not safe. It is why a mid-sized player can farm the edges of a fight and arrive at the whistle in the places, while the monster that ate everyone gets cornered at minute nine.




| MASS | RADIUS | SPEED | VS. START | REJOIN AFTER A SPLIT |
|---|---|---|---|---|
| 20 | 18 | 497 | 100% | 20s |
| 100 | 40 | 338 | 68% | 21s |
| 200 | 57 | 286 | 58% | 23s |
| 400 | 80 | 242 | 49% | 27s |
| 800 | 113 | 205 | 41% | 35s |
| 1600 | 160 | 174 | 35% | 50s |

## 3. Size above a point is a leak

Any cell over **260** mass shrinks continuously, at 0.5% a second. At 400 that is 2 mass a second; at 1,600 it is 8, and standing still is going backwards. Big players have to keep eating purely to stay big, which is what gives the pack behind them a chance.

# The board

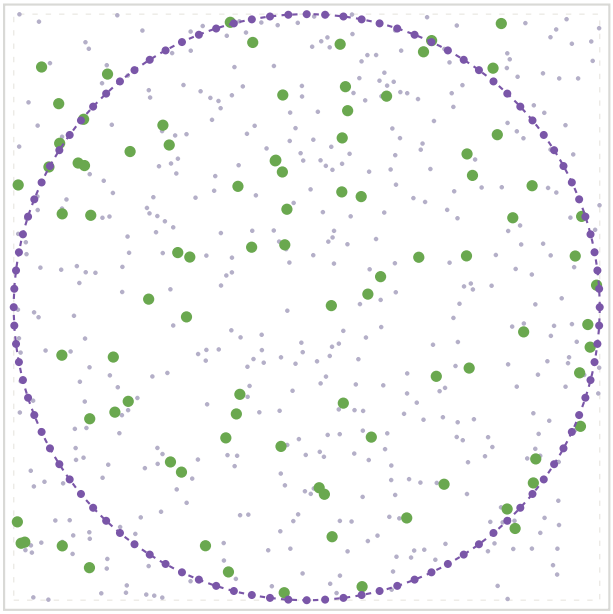
Where you start, and where everything else is. None of it is decorative - the layout decides what your first two minutes can possibly look like.

## Spawn position

At the start of a round the whole field is dealt onto **one ring at equal angular spacing**. Every player gets an identical opening: the same distance to either neighbour, the same distance to the centre, the same distance to the wall. Nobody is handed a quiet corner and nobody opens inside somebody's mouth.

The ring wants to seat everyone 520 units apart, but a full lobby asks for a bigger circle than the arena holds, so it clamps to the widest ring that fits. At a full Standard lobby that means a radius of **4260**, neighbours **268 units** apart, and the ring sitting exactly on the wall padding.

ONE REAL ARENA, TO SCALE



- orbs (one in nine drawn)
- viruses, all 90 of them
- the 100 opening positions

**So a full lobby opens on the perimeter**

With 100 players the ring is pinned 140 units off the wall and the entire middle of the arena is empty. Everyone starts with their back to a wall and open board in front of them. The first thing worth doing is moving inward - the centre is the only part of the map nobody is standing in, and it is where the orbs are untouched.

| AT A FULL LOBBY | STANDARD | HIGH STAKES |
|---|---|---|
| Players on the ring | 100 | 50 |
| Ring radius | 4260 | 2960 |
| Gap to each neighbour | 268 | 372 |
| Clear of the wall | 140 | 140 |

Three details worth knowing. The field is **held on the ring for 3 seconds** before the round begins: aim all you like, but nobody moves, splits or throws until the count runs out, and then everyone goes at once, towards wherever they were pointing. The ring is **turned by a fresh angle every round**, so there is no memorised opening. And the ring is **the only way onto the board**: nobody joins a round once it has started, and nobody respawns in one. Arrive late, or get eaten, and you wait in the lobby for the next round's whistle.

## How the orbs are spread

**Evenly, by Poisson-disc sampling.** Each orb is dropped at a random point, but only where no other orb lies within 86 units of it. The board still looks random - there is no grid to learn - but it cannot clump. There are no rich patches to hunt for and no bare stretches to cross.

| ORBS | STANDARD | HIGH STAKES |
|---|---|---|
| Orbs in play | 4,100 | 2,035 |
| Board area each | 18,888 | 18,889 |
| Never closer than | 86 units | 85 units |
| Typical gap to the nearest | 99 units | 98 units |
| Busiest hundredth of the board | 47 orbs | 25 orbs |
| Emptiest hundredth | 35 orbs | 15 orbs |

### What that ratio means

The busiest hundredth of a Standard board carries only **1.3 times** what the emptiest does. Where you open does not decide how fast you can grow: every stretch of untouched board is worth about the same, so a good first two minutes comes from how cleanly you sweep, not from finding the good patch.

Both modes are tuned to the same density - about 18,888 units of board per orb - so a smaller arena is not a poorer one.

Orbs are replaced the moment they are eaten, so the arena never runs down: the count is held at 4,100 all round. **A replacement follows the same rule**: it can only land where there is room for one, so it fills a gap rather than piling onto ground that is already full, and it never lands under a cell - nobody is handed orbs for standing still. The gap it fills can be anywhere on the board, so a stretch you have grazed stays thin for a while; but nowhere ever fills up past its share.

Ejected mass lands in the same pool. A blob somebody threw is an orb like any other once it comes to rest, worth several times a natural one, and free to anyone who reaches it first.

## How the viruses are spread

Uniformly at random, so unlike the orbs they do clump - and for viruses that matters, because the gaps are wide enough to plan around and the clusters are dangerous.

| VIRUSES | STANDARD | HIGH STAKES |
|---|---|---|
| In the arena at the start | 90 | 45 |
| Board area each | 860,444 | 854,222 |
| Typical gap to the nearest other | 403 units | 424 units |
| Closest pair on a sample board | 13 units apart | 86 units apart |
| Ceiling once players start feeding them | 135 | 68 |

- A virus is 84 units across the middle, and a typical one sits about 403 units from its nearest neighbour - so most of the board is open, and you can usually plan a route that never comes near one.
- But pairs happen. The closest two on the sample board above are 13 units apart, centre to centre - closer than one is wide, so they overlap. A pair like that is a wall if you are over 126 mass, and a fortress if you are under it.
- The population is not fixed once play starts: feeding splits viruses and adds new ones, up to the ceiling. By the endgame there are more viruses than there were at the whistle, and they are wherever players wanted them.

# Splitting

Split and every cell you own halves and fires forward. It is the only way to close distance on something faster than you, and it is the most common way players die.

### What it buys

Reach. A launched half travels a few hundred units instantly, which is how you catch a cell that would otherwise outrun you. It also doubles your cell count, up to a ceiling of 16.

### What it costs

Everything. Each half is half your mass, so each is eatable by things that could not touch you a second ago. And the pieces cannot rejoin for a cooldown that grows with their size - **20 seconds** at small sizes, **50 at mass 1,600**. You are at your most vulnerable for exactly as long as you are at your biggest.

> **The question to ask before every split**
>
> Not 'can I reach it' but 'what eats the halves'. Look at the board behind your target first. If anything near it is bigger than half of you, the split is a donation.

# Ejecting

Eject throws a blob forward. It costs 16 and delivers 13, so about **19% is lost every throw**. You cannot pick your own mass back up for 0.6 seconds, which means recovering it is a decision - turning round and going back - rather than an accident.

There is exactly one thing worth spending that mass on, and it is on the next page.

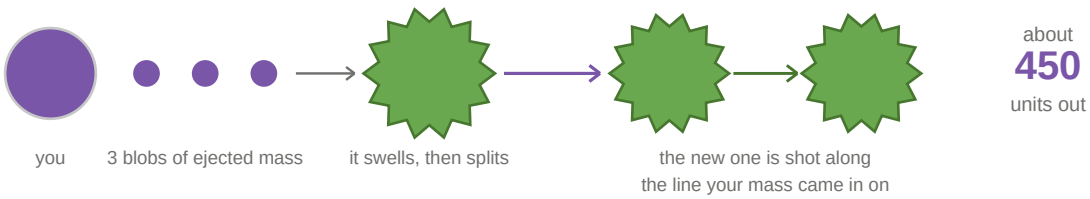
# Viruses: the hazard, and the weapon

The green spiked circles. There are 90 of them in a Standard arena, and they behave completely differently depending on how big you are.

| IF YOU ARE... | A VIRUS IS... |
|---|---|
| **under 126 mass** | Cover. You pass straight over it, and anything big enough to eat you cannot follow you in. Sitting on a virus is the single best defensive move a small cell has. |
| **over 126 mass** | A mine. Touch one and you burst into 9 pieces, scattered and unable to rejoin for a long cooldown, in front of everyone who was chasing you. This is how most large players actually die. |

## Feeding a virus

Shoot ejected mass into a virus and it swells visibly. The **3rd hit splits it**: a new virus is launched along exactly the line your mass came in on, and coasts about 450 units before settling.




This is the only aimed weapon in the game. You are not throwing the virus at someone so much as choosing where you stand while you feed it - the line from you, through the virus, is where the new one goes. Used properly it puts a mine in the path of a player far too big for you to fight, or cuts off the open water a leader was running for.

### What it costs you

Three throws is 48 mass off your own total, gone. Feeding is a pure sink: the virus keeps it, you get nothing back. What you buy is position.

### Two things that will catch you out

Only mass *still in flight* feeds a virus - a blob that runs out of travel short of it has missed. And if you are over 126 mass, walking close enough to aim comfortably is how you burst yourself. Feed from range, or feed while small.

# What is actually at stake

Skip this page if you are playing Practice - nothing below applies. In a wagered round the money is server-side and moves only in response to things that happen in the arena.

| WHEN THIS HAPPENS | YOUR MONEY DOES THIS |
|---|---|
| You join a wagered round | Your stake moves out of your balance and into escrow - your pot. You cannot spend it while the round runs. |
| You eat a staked player | All of their mass becomes yours, and their pot moves into yours. The mass is what you are paid on. |
| You are eaten by a staked player | Your whole pot becomes theirs. |
| You finish in the top 5, alive | You are paid the value of the mass you finished on, 0.005 USDC a point: the mass value on your HUD at the whistle. Your pot goes to the house. Below mass 200 that is less than a 1.00 stake. |
| You finish alive but outside the top 5 | Your pot is forfeited. Surviving is necessary; it is not sufficient. |
| You disconnect and are not eaten | Your body stays in the arena for 6 seconds. Survive that window unclaimed and your stake is refunded. |

**Death ends your round**

Once you are eaten you are out until the next round. You can watch the rest from the spectator view, but there is no respawn and no way back onto the board until the next whistle. Your pot left with your killer. **Your wager is decided the moment you are eaten, not at the whistle.**

## What this means for how you play

- **Mass sets your rank and your payout.** Place in the top 5 and the mass value on your HUD at the whistle is what you are paid. Mass 200 gets a 1.00 stake back; every point above it is profit.
- **Eating a big player is the fastest money there is.** All of their mass becomes yours in one bite, and every point of it is paid if you place.
- **Size makes you a target.** Late in a round the biggest players are carrying the most money, and everybody can see how big they are.

# Levels, rank and matchmaking

Live rounds leave you with two numbers that last from one game to the next. **XP** counts what you have won and only ever goes up. Your **rating** measures how well you play, and goes up and down. Neither is money, and Practice against bots earns neither.

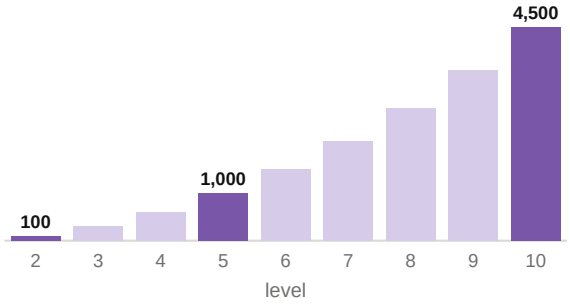
## Experience: winning, counted

You earn XP for one thing: **being alive at the whistle in the paid places** - the same line the payout draws, with places counted among everyone still standing, exactly as the payout counts them.

| FINISH | XP |
|---|---|
| 1st | +100 |
| 2nd | +70 |
| 3rd | +50 |
| 4th | +35 |
| 5th | +25 |

Eaten, or alive outside the top 5: no XP.

TOTAL XP TO REACH EACH LEVEL




Each level asks for 100 XP more than the one before. The marked bars are levels 2, 5 and 10.

## Levels

Your level is your total XP on a curve: reaching level *L* takes 50 × *L* × (*L* - 1) XP. One first place takes you to level 2, about 10 first places to level 5, and about 45 to level 10. There is no top level. A level changes nothing about how the game plays: it is a record of what you have won.

| LEVEL | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 15 | 20 | 30 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Total XP | 100 | 300 | 600 | 1,000 | 1,500 | 2,100 | 2,800 | 3,600 | 4,500 | 10,500 | 19,000 | 43,500 |

## Your rating

Everyone starts at 1,000. After every live round you were dealt into, your rating is settled against each other person in that round: finish above someone and you beat them, finish below and they beat you. What each result is worth depends on the gap between your two ratings, so finishing above a stronger player moves you much further than finishing above a weaker one.

| RANK | RATING |
|---|---|
| Bronze | below 900 |
| Silver | 900 - 1,099 |
| Gold | 1,100 - 1,299 |
| Platinum | 1,300 - 1,499 |
| Diamond | 1,500 and up |

A new account starts in Silver.

- **One round moves you by at most 24**, however many played. For your first 10 rated rounds the most is 48, so a new rating finds its level quickly - the menu marks it *provisional* until then.
- Beat a player rated 400 above you, one on one, and you gain **22**. Beat one rated 400 below you and you gain **2**.
- The finishing order is survivors by their place at the whistle, then everyone eaten, the last one out ranked highest. Only people are rated: bots in the room do not move your rating.

## Matchmaking

You choose a stake, never a room: the matchmaker seats you. Each room is described by the average rating of the players in it, and takes players within a window of that average - **±200** to start with, widening by **10** for every second its lobby has been waiting for a round.

| LOBBY HAS WAITED | TAKES RATINGS WITHIN |
|---|---|
| just opened | ±200 |
| 15 seconds | ±350 |
| 30 seconds | ±500 |
| 1 minute | ±800 |
| 2 minutes | ±1,400 |

### Where you are seated

- **A room you fit.** Between rounds before mid-round, then the fuller one, so lobbies fill rather than scatter.
- **Otherwise a new room**, which takes its rating from you.
- **Otherwise the room nearest your rating.** Your rating decides where you sit, never whether you get a seat.

The window is why a quiet evening still plays. A busy server fills lobbies fast and keeps them tight, so players are sorted by skill. A lobby that is waiting a long time for enough players soon takes almost anybody - after a minute, anyone within 800 of its average - rather than splitting a small crowd into rooms that never start.

## What it means for how you play

- **One life per round.** Eaten, you are out until the next round: there is no respawn to play on with.
- **Leaving does not dodge a loss.** Walk away mid-round and you are out the moment your cell leaves the arena, ranked with the eaten.
- **Surviving is worth rating even outside the top 5.** Everyone alive at the whistle finishes above everyone who was eaten.
- **Where to see it.** The menu and the lobby card show your level, the XP to your next one, and your rank. The lobby card says what the round was worth: *+100 XP · Level 2! · Rating +24*.

> **Bots in staked rooms**
>
> Live rounds on the real server are players only. Test servers fill staked rooms with bots; there, the difficulty you pick also chooses your room, and you are seated only with players who picked the same.

# Playing against bots

Practice pits you against bots in your own browser. Pick **Easy**, **Normal** or **Hard** under the Practice tier in the menu - Normal is the default. It is remembered, takes effect at once even mid-run, and Practice earns no XP or rating.

| | EASY | NORMAL | HARD |
|---|---|---|---|
| Goes after smaller cells within | 430 units | 500 units | 620 units - all it can see |
| Hunts cells it cannot see | no | no | up to 1,800 away, when nobody else is near it |
| Eats orbs | the nearest | the nearest | best value for distance, thrown mass first |
| Leads a moving target | no | no | aims 0.3s ahead |
| Splits onto smaller cells | never | close range only, every 8s | from 90% of full reach, every 1.5s, even when already in up to 4 pieces |
| Double split | never | never | yes - a quarter thrown from the half in flight |
| Shoots viruses at bigger cells | never | only one it is already lined up behind | spots targets up to 1,000 away and walks up to 520 to line a virus up |
| Hides behind viruses | never | never | yes, within 400 - the chaser bursts on it |

Every level runs from anything that can eat it. No level splits while something nearby could eat one of its halves.

## How close is too close

A bot splits onto you when half of it still eats you and the throw lands. These are the distances, centre to centre in world units, from which it can: the smaller you are compared with the bot, the further they reach. For scale, a small cell sees about 760 units in each direction.

| BOT MASS | NORMAL SPLITS IF YOU ARE | HARD SPLITS IF YOU ARE | HARD DOUBLE-SPLITS IF YOU ARE |
|---|---|---|---|
| 100 | under 40, within 174 | under 44, within 260 | under 22, within 352 |
| 200 | under 79, within 197 | under 88, within 295 | under 44, within 394 |
| 400 | under 158, within 231 | under 177, within 345 | under 88, within 452 |
| 800 | under 317, within 279 | under 354, within 416 | under 177, within 534 |

## Beating Hard bots

- **They hunt when they are alone.** A Hard bot with nobody else in sight comes looking for you from up to 1,800 away. Near other cells it fights where it is, so company is cover.
- **Respect the double split.** It lands a quarter of the bot about a third further than a single split reaches. Read the table above before you drift near anything twice your size.
- **Turn late.** Hard bots aim 0.3s ahead of where you are going. A change of direction as they commit sends the split into empty space.
- **Above mass 126, mind the viruses.** Never chase a small Hard bot through one: it runs behind a virus so that you burst on it. And a virus near a big Hard bot that swells as blobs hit it is being aimed at you - it takes 3 feeds, and the new virus flies along the line the blobs came in on.
- **Stay near something bigger than the bot.** Bots of every level refuse to split where something could eat their halves.

# The eight skills

Rounds are not won by reflexes. They are won by eight separate habits, each of which can be practised on its own. In rough order of how much a beginner gains from fixing it.

1 SURVIVAL

## Threat arithmetic

At any moment you should know, without thinking, which cells on your screen can eat you and which you can eat. The line is 9.8%: anything over 1.098 times your mass is lethal, anything under 0.911 times is food.

Beginners read size roughly and get it wrong at the margin, which is exactly where rounds are decided. The margin is narrow on purpose - a cell 5% bigger than you is not a threat, and a cell 10% bigger is.

**Drill.** In Practice, park next to bots of similar size and read their mass off the leaderboard before you commit. Get used to the visual difference a 10% mass gap makes - it is smaller than you expect, because radius grows with the square root of mass.

2 SURVIVAL

## Split discipline

The commonest way to lose a round you were winning. A split halves every cell you own and locks the pieces apart for 20 to 50 seconds.

Good players split when the target is certain and the surrounding board is empty. Bad players split at anything that looks catchable, and spend the next 30 seconds as eight easy meals.

**Drill.** Play a whole Practice round without splitting once. You will finish smaller but you will learn how much of the board you can hold without it. Then add splits back one at a time, only when nothing bigger than half your mass is within a screen.

3 BOTH

## Virus literacy

Know where every virus near you is, and which side of 126 mass you are on. Under it they are your best cover - a big chaser physically cannot follow you onto one. Over it they are the thing most likely to kill you.

The advanced half is offensive: feeding one takes 3 throws and lets you place a new virus roughly 450 units away along a line you choose.

**Drill.** Grow past the threshold in Practice and deliberately run a bot chase near viruses until you can feel the danger radius. Then spend a round doing nothing but feeding viruses, to learn the range and lead of a throw.

4 GROWTH

## Orb economy

Orbs are 4 mass each and there are 4,100 of them. Early on, they are the whole game: you need roughly 20 to reach 100 from your opening 20.

The skill is pathing. Orbs are evenly spaced, so the ground worth sweeping is the ground nobody has grazed yet: take it in straight lines rather than chasing individual orbs, and do it in the quiet parts of the arena while the aggressive players trade early deaths.

**Drill.** Time yourself from spawn to mass 100 in Practice. A good line beats a greedy one; compare a straight sweep against darting at whatever is nearest.

5

BOTH

## Speed awareness

Every mass you gain makes you slower, on a curve that never flattens. By mass 400 you are at roughly half your opening speed; by 1,600, a third.

Two consequences people miss. You cannot catch anyone much smaller than you without splitting, so growth past a point converts into pure vulnerability. And when you are the biggest thing on the board, running away has stopped being an option - you have to fight or use viruses.

**Drill.** Grow to 800 in Practice and try to catch a starting-size bot. You cannot. Learn that before a real round teaches you.

6

SURVIVAL

## Map and wall sense

The arena is square and the walls do not push back. A cell pinned against one has lost half its escape directions, and experienced players herd others into corners deliberately.

Keep the middle of the board available to you. The minimap exists for exactly one question: how much room is behind me?

**Drill.** Watch your minimap dot rather than the arena for thirty seconds in Practice. Notice how often you drift into an edge without deciding to.

7

WINNING

## Clock management

The round is 10 minutes and only the last moment of it counts. Rank at minute four is worth nothing; rank at 0:00 is worth everything.

Early round, take risks - a death at minute two costs you a stake you can re-enter with next round. Late round, with a place in hand, every fight you do not need is a fight that can only cost you. The last 30 seconds are for defending a position, not improving it.

**Drill.** In the final minute, find the line under 5th on the leaderboard and play only against the players either side of it. Everything else on the board is irrelevant to you.

8

WINNING

## Pot awareness

Only in wagered rounds, and it is what separates a good player from a profitable one. A paid place is paid the value of its mass at the whistle, and mass 200 only gets your stake back.

So a round spent creeping to a safe fourth place at mass 150 pays less than you staked. A round in which you ate two big players and finished fifth on mass 900 pays four and a half stakes. Rank decides whether you are paid; mass decides how much.

**Drill.** After each wagered round, ask which of your fights grew your mass and which merely protected your rank. Most players find they spent the round on the second kind.

# Technique

The skills chapter is what to work on. This is how the work is actually done - eight techniques, and the numbers each one turns on.

## Movement and positioning

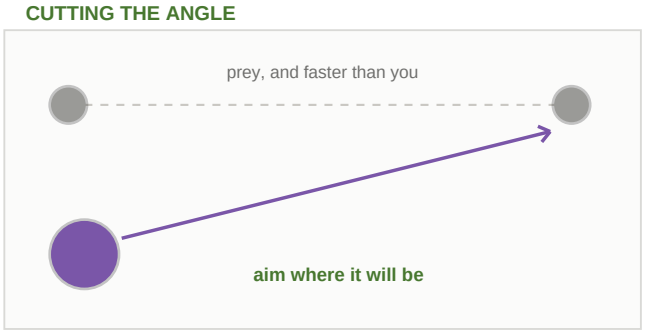
Avoiding larger cells while creating angles on smaller players.

Start from the fact that makes every chase futile: **every cell you are big enough to eat is faster than you are**. Speed falls with mass, so being able to eat something guarantees it can outrun you. Even the marginal target - the one only 9.8% smaller - has the legs on you.

| YOU ARE MASS 400 AND THEY ARE | THEIR SPEED | YOURS | THEIR ADVANTAGE |
|---|---|---|---|
| 364 (the biggest you can eat, just) | 248 | 242 | 2.3% faster |
| 308 | 258 | 242 | 6.5% faster |
| 200 | 286 | 242 | 18.1% faster |
| 100 | 338 | 242 | 39.5% faster |
| 50 | 399 | 242 | 64.7% faster |

So pursuit is not a tactic, it is a way of losing ground slowly. What works is geometry: take the inside line, aim at where the target must go rather than where it is, and push it toward something that limits its options - a wall, a virus, or a bigger player.




Against larger cells the same geometry runs backwards. Keep two escape directions open at all times; the moment you have one, you are being herded. Open water behind you is worth more than orbs in front of you.

**The tell you got it wrong.** You spent ten seconds behind the same player and never closed the gap. That was never going to work - you needed an angle, a wall, or a split, from the first second.

## Split timing and accuracy

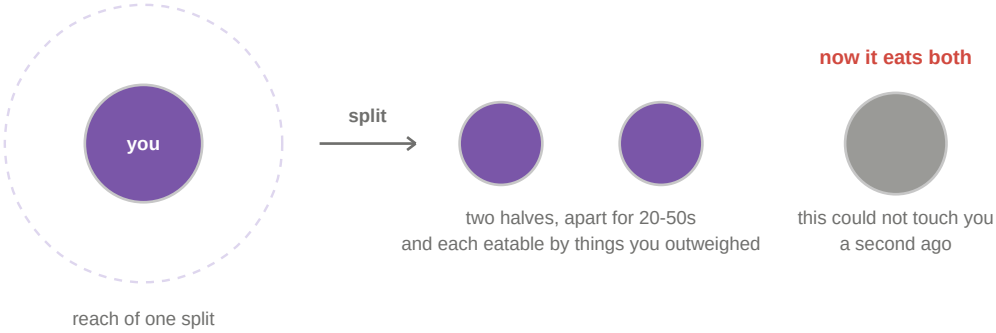
Judging whether a split will reach and consume another player without leaving you vulnerable.

The split is the only thing that closes distance, because the launch is roughly **5.6 times your cruising speed**. It is also a timer on your own vulnerability. Both halves of the decision are in one table.

| YOUR MASS | EACH HALF | REACH FROM YOUR CENTRE | YOU STAY IN PIECES FOR |
|---|---|---|---|
| 60 | 30 | about 282 units | 20s |
| 100 | 50 | about 308 units | 21s |
| 200 | 100 | about 355 units | 23s |
| 400 | 200 | about 422 units | 27s |
| 800 | 400 | about 517 units | 35s |
| 1600 | 800 | about 651 units | 50s |

### Three questions, in order

- **Will it reach?** Read the table, not your instinct. At mass 200 you throw about 355 units; at 800, about 517. Anything beyond that is a donation.
- **Will the half be big enough?** The piece that arrives is half your mass, and it still needs to be 1.098x the target. Halving 200 gives you 100, which eats nothing above 91.
- **What eats the halves?** The question people skip. Look behind the target before you commit, not after.




**The tell you got it wrong.** You landed the split, ate the target, and were eaten yourself before the pieces rejoined. The split worked and the decision did not.

## Mass management

Deciding when to split, merge, feed, or preserve mass.

You have four things you can do with mass, and each is a trade you should be able to price.

| DO THIS | YOU PAY | YOU GET |
|---|---|---|
| Split | Half your size per cell, and 27s apart at mass 400 | Reach, and a second mouth |
| Merge | Nothing, but you cannot hurry it | Back to one fast, safe body |
| Feed a virus | 16 mass a throw, 19% of it lost outright | A hazard placed where you choose |
| Preserve | Nothing below 260 mass; above it you leak 0.5% a second | Speed, and the ability to run |

The non-obvious one is **preserve**. Because speed falls with mass and everything above 260 decays, there is a size past which extra mass makes you worse at everything except eating people who cannot escape. Growing beyond it is a decision, not a reward - take it when you intend to hunt, not because the orbs were there.

> **Mass in pieces is not mass**
>
> Sixteen cells of 50 is not a 800-mass player. It is sixteen 50-mass players who happen to share a name, each of which can be eaten separately, and none of which can rejoin for 35 seconds. Count your effective size as your *largest cell*, not your total.

## Risk assessment

Deciding when a target is worth pursuing versus when to retreat.

Price the attempt before you start it. A pursuit costs time, position and usually a split; what it returns is all of the target's mass, plus - in a wagered round - their entire pot.

- **Worth it:** a staked player carrying pots, cornered, with nothing bigger than half your mass within a screen.
- **Not worth it:** anything in open water that can see you coming. It is faster than you and the chase is free for it.
- **Never:** a target sitting on or beside a virus while you are over 126 mass. That is the trap, and good players set it deliberately.

## Map awareness

Tracking nearby threats, viruses, escape routes and crowded areas.

Four things to keep a running count of, in this order of urgency: **what can eat me**, **where the viruses are**, **which way is open**, and **where the crowd is**.

The crowd matters more than beginners expect. A fight between two large players generates fragments, ejected mass and burst cells - a burst player scatters into 9 pieces - and the profit is in arriving after it, not during. Equally, a quiet quadrant is where you grow unmolested for the first two minutes.

How far you can see is not a matter of opinion. Everyone is shown the same **760 units** in every direction, and that does not change until you pass about **446 mass** - past which your view finally starts growing with your size.

Two things follow. For most of a round every player on the board has exactly your horizon, so anything you cannot see cannot see you either. And the handful of players large enough to have outgrown the floor are the only ones with an information advantage - which is part of why a leader is hard to ambush.

**The tell you got it wrong.** You died to something you never saw. Nine times in ten that is a wall at your back, not bad luck.

## Prediction

Anticipating opponents' movement and split attacks.

Two things to read.

**Movement.** Everyone else is drawn from the last thing the server sent, so on a slow connection you are seeing them a fraction of a second in the past. Lead your targets. This cuts the other way too: your own cell is predicted locally and is exactly where you think it is, which is why your escapes work better than your chases.

**Split attacks.** A player about to split lines up first - they stop cutting corners and start travelling straight at you. The warning is short and the reach is long (422 units at mass 400), so the counter is distance held in advance, not reaction. If something twice your size is pointing at you and closing, you are already inside its range.

> **The counter nobody uses**
>
> A player who has just split is two half-sized cells that cannot rejoin for tens of seconds. If they miss you, they are prey - and for that window they are the most profitable target on the board. Baiting a split and turning on the pieces is the highest-value play in the game.

## Virus mechanics

Skilled players use viruses offensively and defensively.

**Defensively**, while you are under 126 mass, a virus is a place nothing dangerous can follow you. Learn where they are and treat them as safe squares - a chase that would kill you in open water ends the moment you sit on one.

**Offensively**, there are two plays. The first needs no setup: herd a large player toward a virus. They know the danger, so the threat of it alone steers them, which is often all you want.

The second is feeding. 3 blobs of ejected mass split a virus and launch a new one along the line your mass came in on, about 450 units. That is how a small player removes a large one without ever being big enough to fight it - and how you close off the open water someone was running for.

- Cost: 48 of your own mass, unrecoverable.
- Only mass still in flight counts, so feed from a distance you can actually throw - a blob that stops short has missed.
- Do not set up a feed while over 126 mass without watching your own drift. Walking in to aim is how people burst themselves.

## Team tactics

Feeding, baiting, splitting and coordinating - where applicable.

The mechanics exist, so it is worth knowing what they are. **Feeding** transfers mass to another player at 19% loss per throw. **Baiting** means one player presenting as catchable so a third party splits at them, and the partner taking the pieces. **Coordinated splitting** means two players cutting off both escape directions at once, which defeats the geometry in the positioning section above.

> **Where this applies, and where it does not**
>
> In Practice, and in any friendly round, all of the above is fair play and good fun. **In a wagered round it is not a tactic, it is collusion.** Two accounts moving stakes between themselves by feeding is precisely the behaviour a staked game has to exclude, and the fact that the mechanics allow it is an open problem rather than a licence. Treat wagered rounds as what they are: every other player is an opponent.

The one piece of it that is always legitimate is reading other people's coordination. If two cells keep arriving together, assume they are working together and stop treating either as a lone target.

# A ten-minute playbook

One way to structure a round. Not the only way, but a shape to depart from rather than playing on instinct for ten minutes.

| CLOCK | PRIORITY | WHAT YOU ARE DOING |
|---|---|---|
| 10:00 - 8:00 | Grow, quietly | You open equidistant from two neighbours who are exactly your size, so nobody can eat anybody. Leave. Sweep orbs away from the ring and let the players who fight at the opening whistle eat each other. |
| 8:00 - 5:00 | Convert size into kills | You should be comfortably over 100. Hunt players in the band you can actually eat - anything under 0.91x your mass - and take them without splitting where possible. In a wagered round this is where the payout is made: a kill is mass you could not farm. |
| 5:00 - 2:00 | Manage the crown | If you are large, you are slow and you are decaying above 260. Stay off viruses, stay off walls, and keep eating enough to cover the leak. If you are mid-sized, pick off the fragments from other people's fights - a burst player is nine free meals. |
| 2:00 - 0:45 | Find the line, then the zone | Look at where 5th place sits. If you are above it, your job is no longer to grow. If you are below it, you need one player, and viruses are how a smaller cell removes a bigger one. At 1:40 the zone is marked on your minimap: if you are outside it, go now, while the board is still wide. |
| 0:45 - 0:00 | Survive | Do not split. Do not chase. Do not go near a virus. Everyone left is in the same square now, so open water is scarce: hold what you have and let the clock run out. |

## The five mistakes that cost the most

- **Splitting to catch something you did not need.** The halves are what gets eaten, and the cooldown is measured in tens of seconds.
- **Carrying more than 260 mass without a plan.** Above it you leak, you are slow, and viruses are lethal to you.
- **Chasing a rank instead of mass.** A place on small mass pays back less than the stake.
- **Fighting in the last thirty seconds.** A place in hand is worth more than a place you might win.
- **Counting on a second chance.** There is none. Get eaten and you sit out the rest of the round, and your mass and your pot have already gone to the player who ate you.

## The closing zone

When the clock reads 1:30 the board starts closing in to a square 60% of its width, placed at random. You are warned 10 seconds before it moves: a banner counts down and the square appears on your minimap. Its edges reach the square over 45 seconds and hold there until the whistle. Outside it each cell loses 10% of its mass a second plus 2, and is gone below 10: a starting cell lasts 3 seconds out there, and a 1,000-mass cell halves in 7.

- **You can always make it.** The edges move slower than you can, so heading in when told is enough. Waiting is what kills.
- **The food moves in with it.** While the zone closes, new orbs only appear inside it, never where they would cost you mass to collect.
- **A zone death is nobody's kill.** Your stake is lost, as it would be to a bot, and whoever bit you earlier is paid nothing for it.

# Quick reference

Every number the game runs on, on one page.

## Eating and growing

| | |
|---|---|
| Starting mass | 20 |
| Orb value | 4 mass |
| To eat a rival | you must be 1.098x their mass (9.8% bigger) |
| Reward for eating | exactly the mass you swallow |
| Decay | above 260 mass, 0.5% per second |
| Speed | falls as mass$^{-0.24}$ |

## Splitting and ejecting

| | |
|---|---|
| Maximum cells | 16 |
| Rejoin cooldown | about 20s at small sizes, 50s at mass 1,600 |
| Eject cost | 16 mass spent, 13 delivered (19% lost) |
| Eject range | about 237 units past your own membrane |
| Own-mass cooldown | 0.6s before eating your own throw back |

## Viruses

| | |
|---|---|
| Virus mass | 110 |
| Dangerous to you above | 126 mass |
| Pieces you burst into | 9 |
| Feeds to split one | 3 blobs of ejected mass, still in flight |
| Where the new one goes | along the line your mass came in on, about 450 units |
| Population ceiling | 1.5x the count the arena seeds |

## The round

| | STD | HIGH |
|---|---|---|
| Stake | 1.00 | 2.00 |
| Players needed | 100 | 50 |
| Arena | 8,800 | 6,200 |
| Orbs | 4,100 | 2,035 |
| Viruses | 90 | 45 |
| Round length | 10:00 | 10:00 |
| Zone closes | 1:30 left | 1:30 left |
| Final square | 5,280 | 3,720 |
| Paid places | 5 | 5 |
| Countdown | 5s | 5s |
| Hold on the ring | 3s | 3s |
| Standings | 5s | 5s |

Practice is a third tier with no stake, no lobby and no server: it runs against bots in your own browser, at the difficulty you choose. Use it for everything in the skills chapter.

## Progress

| | |
|---|---|
| XP for a win | 1st +100 / 2nd +70 / 3rd +50 / 4th +35 / 5th +25 |
| Level L needs | 50 x L x (L - 1) XP in total |
| Rating | starts at 1,000; moves at most 24 a round (48 for the first 10) |
| Ranks | Bronze <900, Silver 900+, Gold 1,100+, Platinum 1,300+, Diamond 1,500+ |
| Matchmaking | rooms take ratings within ±200, widening 10 a second while the lobby waits |